DESHAN FERNANDO

Software Engineering Intern | API Management & Backend

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Portfolio: https://deshanfdo.github.io/Portfolio/

PROFESSIONAL SUMMARY

Software engineering undergraduate with internship experience building secure, production-oriented web applications. Skilled in REST API design, API gateway patterns, authentication/authorization (JWT, RBAC), and working with SQL/NoSQL databases. I am interested in opportunities to contribute to scalable enterprise systems, especially API management and integration platforms while strengthening backend engineering and cloud-native development skills.

EDUCATION

BSc (Hons) in Computer Science (Reading) | University of Westminster (UK) / Informatics Institute of Technology (Sri Lanka)
Jan 2023 – Present

Foundation Certificate in Higher Education — IT (Distinction) | Informatics Institute of Technology
Jan 2023 – Sep 2023

EXPERIENCE

Software Engineer Intern, **Sri Lanka Telecom**
Jun 2025 – Present

- Built and maintained RESTful backend services in Node.js and TypeScript to manage assessments, attempts, and evaluation workflows.
- Implemented authorization patterns including RBAC and panel-level data isolation to support secure multi-tenant data handling.
- Developed modular React components and improved assessor workflows through refactoring and performance-focused UI changes.
- Optimized API behavior and database interactions to improve platform efficiency and reliability in a production-grade environment.
- Contributed in an Agile workflow through code reviews, debugging, and iterative refactoring of existing modules.

PROJECTS

Enterprise Developer API Portal

- Built a full-stack real-time API dashboard aggregating Weather, Stock Market, and Cryptocurrency data from multiple external APIs.
- Implemented enterprise API management patterns using WSO2 API Manager (Docker) as an API gateway with JWT validation (toggleable via env var) and OpenAPI 3.0 specs.
- Designed versioned REST endpoints (/api/v1/*) with MVC layering (controllers/services), centralized error handling, request logging, and CORS for cross-origin access.
- Automated CI/CD with GitHub Actions and deployed full-stack to Render plus a static GitHub Pages demo with offline/demo-data fallback.

- Implemented resilience patterns (multi-API fallback for weather + server-side fallback data) to keep demos usable when upstream APIs fail.
- **Tech:** React, Vite, Node.js, Express, WSO2 API Manager, Docker Compose, GitHub Actions | **GitHub:** github.com/deshanFdo/apiPortal

LawLinkLK: Client & Lawyer Connecting Platform

- Built a full-stack web app enabling secure messaging and appointment scheduling with role-based access.
- Implemented authentication and real-time communication using Socket.IO; added an AI-powered assistant for user support.
- **Tech:** React, Node.js, MongoDB, Tailwind CSS, Socket.IO | **Link:** lawlinklk.com

Smart Employee System

- Developed an enterprise-style application featuring REST APIs, JWT authentication, and role-based authorization.
- Integrated AI chatbot capabilities to improve the end-user experience.
- **Tech:** Node.js, TypeScript, React, MySQL

AI Floorplan-to-3D House Generator

- Built an end-to-end floorplan-to-3D pipeline that converts 2D floorplan images into structured geometry (rooms/walls/doors/windows) and 3D-ready outputs.
- Designed a modular multi-stage CV/ML architecture (detection → segmentation → geometric post-processing → generation) with intermediate artifacts for debugging and reproducibility.
- Implemented a Python backend API to orchestrate inference, validate uploads, cache results, and export JSON + visual overlays; built a React (Vite) UI to run jobs and visualize outputs.
- **Tech:** Python, Ultralytics YOLO, PyTorch, OpenCV-style processing, React, Vite

TECHNICAL SKILLS

Languages: Java, JavaScript, TypeScript, Python, SQL, C#
Frontend: React, HTML5, CSS3, Tailwind CSS, Material UI, Axios
Backend: Node.js, Express.js, REST APIs, JWT, RBAC, MVC
API Management: WSO2 API Manager, OpenAPI 3.0
Databases: MySQL, MongoDB
Tools/Practices: Git, GitHub, Postman, Jira, Linux basics, Agile/Scrum, OOP, Clean Code, Debugging, Unit testing basics
Cloud/Deploy: Azure Fundamentals, Vercel, Render, Docker, CI/CD basics

CERTIFICATIONS

- Postman API Fundamentals — Student Expert
- Introduction to Databases for Back-End Development — Meta (Coursera)
- Python for Data Analysis: Solve Real-World Challenges — LinkedIn Learning
- Microsoft Azure Fundamentals (AZ-900) — LinkedIn Learning
- Cloud Computing: Understanding Core Concepts — LinkedIn Learning

ACHIEVEMENTS & ACTIVITIES

- Team Leader — IEEE Xtreme Encode (6-week competitive programming challenge); placed 10th in Region 10 (Jul–Sep 2024).
- Participant — Hult Prize On-Campus Competition.